

Fixed

HelixTranslate — Fixed (Closed Workable Items)

Revision: 1 **Last modified:** 2026-06-14T15:37:53Z **Authority:** §11.4.15 (status) · §11.4.16 (type) · §11.4.19 (column-alignment) · §11.4.33 (type-aware closure vocabulary) · §11.4.53 (Fixed_Summary parity) · §11.4.54 (ATM-NNN ticket IDs)
Scope: the closed-archive tracker. Every entry is a real, landed, in-git fix from this session's mutation-proven bug-hunt campaign + the format-matrix / PDF / DeepSeek / WebSocket / SSH work. Each carries a stable [ATM-NNN] id, ****Status:****, ****Type:****, a 1-2 line root cause, and the commit hash as captured evidence (§11.4.123 — no fix is claimed that is not in `git log`).

Sort order: ATM id ascending (allocation order). The companion `Fixed_Summary.md` sorts most-recent-first.

§1. [ATM-001] `splitText` dropped paragraph separators at chunk boundaries causing structural data loss

Status: Fixed (→ `Fixed.md`) **Type:** Bug

- Root cause: large-chapter chunking concatenated chunks without re-inserting the paragraph separator at boundaries, silently flattening structure in the translated output. Mutation-proven.
- Evidence: commit `24b0fd0`

§2. [ATM-002] In-memory translation cache-key collision served the wrong translation

Status: Fixed (→ `Fixed.md`) **Type:** Bug

- Root cause: the in-memory cache key did not distinguish all inputs, so distinct (text,context) tuples collided onto one entry and a prior translation was returned for new content. Mutation-proven.
- Evidence: commit `bd8a1ef`

§3. [ATM-003] FB2 writer dropped deeply-nested (>2 level) section content as translated-text data loss

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the FB2 writer recursion stopped serializing section content beyond two nesting levels, dropping translated text for deeply structured books. Mutation-proven.
- Evidence: commit 71babe1

§4. [ATM-004] Anthropic client dropped all content blocks after the first, truncating translations

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the Anthropic response parser read only content [0], so multi-block responses were truncated or emptied. Mutation-proven.
- Evidence: commit 87a8048

§5. [ATM-005] Distributed ResultCache spurious eviction on update + round-robin index panic after pool shrink

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: cache update path evicted a still-valid entry, and the round-robin worker index was not re-bounded after the pool shrank, causing an out-of-range panic. Mutation-proven.
- Evidence: commit edadd73

§6. [ATM-006] EPUB percent-encoded/relative hrefs dropped chapters and cover; HTML test-hack corrupted content

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the EPUB parser failed to resolve percent-encoded / relative manifest hrefs (losing chapters + cover) and a leftover hardcoded test-hack corrupted HTML content. Mutation-proven.
- Evidence: commit 47ea69d

§7. [ATM-007] Verification parseNote dropped EXAMPLES-terminated notes, produced NaN confidence, under-counted multipass sections

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: note parsing terminated early on EXAMPLES, divided by zero into NaN confidence, and the multipass section counter was off. Mutation-proven.
- Evidence: commit 34e09f8

§8. [ATM-008] Redis makeCacheKey metadata delimiter-injection collision served the wrong translation

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the Redis cache key concatenated metadata with a delimiter that could appear in the data, letting distinct inputs hash to the same key and return a wrong cached translation. Mutation-proven.
- Evidence: commit 9946d85

§9. [ATM-009] EPUB[?]Markdown round-trip destroyed code blocks, lost GFM tables, corrupted backslash-escapes

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the markdown round-trip converter mishandled fenced code blocks, GFM tables, and backslash escapes, losing or corrupting content on the EPUB→MD→EPUB path. Mutation-proven.
- Evidence: commit d055b8c

§10. [ATM-010] API translateText silently dropped script=cyrillic and accepted invalid script; verifier handlers returned wrong HTTP status

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the translate API ignored the cyrillic script flag and did not validate the script value; verifier handlers mapped every GetModel error to one wrong status code. Mutation-proven.
- Evidence: commit b8bc3ec

§11. [ATM-011] Verifier fallback chain randomly shuffled (lost score order); OpenRouter pricing decode failed on string values

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the fallback chain was non-deterministically shuffled, discarding score-priority order, and OpenRouter pricing decoding assumed numeric JSON where the API returns strings. Mutation-proven.
- Evidence: commit 4a95b06

§12. [ATM-012] gRPC GetTranslationStatus returned the wrong error code; cleanupOldSessions leaked a timeout context

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: status lookup returned an incorrect gRPC code for missing sessions, and the session-cleanup loop never cancelled its per-iteration timeout context. Mutation-proven.
- Evidence: commit 45f6152

§13. [ATM-013] CWE-208 username-enumeration timing oracle in AuthenticateUser

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: authentication short-circuited on unknown usernames before the password hash compare, leaking account existence via response timing. Mutation-proven.
- Evidence: commit 5871848

§14. [ATM-014] Preparation failed-chapter analysis misattribution + nested-subsection analysis content loss

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: a failed chapter analysis was attributed to the wrong chapter index and nested subsection content was dropped from the analysis pass. Mutation-proven.
- Evidence: commit 367d88b

§15. [ATM-015] Directory batch to a non-existent output dir collided all files onto one path (data loss)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: batch directory translation did not create the output directory and computed a single colliding output path, so every file overwrote one another. Mutation-proven.
- Evidence: commit 53b5056

§16. [ATM-016] Coordination TranslateWithRetry ignored ctx cancellation; consensus skipped available instances

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the retry loop did not honor context cancellation, and the consensus selector skipped instances that were actually available. Mutation-proven.
- Evidence: commit df8a8cb

§17. [ATM-017] Successful-but-empty translation silently wiped source content (data loss)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: when a provider returned an empty string with a success status, the pipeline wrote the empty result over the source instead of preserving the original. Mutation-proven.
- Evidence: commit 837f528

§18. [ATM-018] Dashboard ETA formatDuration trailing space + non-unique progress event IDs

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: formatDuration emitted a trailing space and progress event IDs were not unique, breaking dashboard event de-duplication. Mutation-proven.
- Evidence: commit b1b747f

§19. [ATM-019] Qwen client posted to native DashScope path under an OpenAI-compatible base (wrong URL + response-shape mismatch)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the Qwen client mixed the native DashScope endpoint with an OpenAI-compatible base URL, producing a wrong request URL and a response shape the parser could not read. Mutation-proven.
- Evidence: commit 556002d

§20. [ATM-020] preparation-translator -providers flag silently ignored; CLI explicit -provider overridden by config

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the -providers flag was parsed but never consumed, and an explicit -provider was overwritten by the config value. Mutation-proven.
- Evidence: commit 5dfe003

§21. [ATM-021] Storage cache dup-tuple served stale translation (added UNIQUE lookup-hash index + idempotent UPSERT)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: duplicate lookup tuples could store divergent rows with no uniqueness constraint, serving a stale translation; fixed with a UNIQUE index, idempotent UPSERT, and migration. Mutation-proven.
- Evidence: commit 367adce

§22. [ATM-022] VersionManager.metrics data race / lost updates (+ §11.4.85 stress/chaos suites)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: concurrent metric updates in the distributed VersionManager raced and lost increments; fixed with synchronization plus stress/chaos coverage. Mutation-proven.
- Evidence: commit bf81517

§23. [ATM-023] Qwen compatible-mode alignment to default base + §11.4.98-honest live test

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: Qwen compatible-mode used an inconsistent base URL; aligned to the default compatible base and made the live test honest per §11.4.98.
- Evidence: commit 2f0e359

§24. [ATM-024] DOCX input non-functional (license-gated unioffice) — rewritten as a stdlib OOXML parser

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: DOCX parsing depended on the license-gated unioffice library and was inert; rewritten as a stdlib OOXML parser, dropping the dependency. Real-artifact proven.
- Evidence: commit d433210

§25. [ATM-025] API-key gate ignored provider env vars (DEEPSEEK_API_KEY etc.) in unified-translator

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the unified CLI API-key gate did not read provider-specific environment variables, blocking otherwise-valid providers. Real DeepSeek E2E proof committed.
- Evidence: commit c6c2930

§26. [ATM-026] DOCX rejected by the CLI pipeline despite a working parser (detector.IsSupported missing DOCX)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the format detector's supported-set omitted DOCX, so the CLI rejected files the parser could handle. Real E2E proven.
- Evidence: commit 6304f60

§27. [ATM-027] FB2 + EPUB input broken end-to-end (convertToMarkdown re-parsed already-extracted text)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the unified pipeline re-ran convertToMarkdown over already-extracted FB2/EPUB text, corrupting the content; fixed so all five input formats translate. Real E2E matrix proven.
- Evidence: commit 41063cb

§28. [ATM-028] Zhipu missing from the API-key env map in unified-translator (+ stale-allowlist audit)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the Zhipu provider was absent from the env-key map, so its key was never picked up. Multi-provider E2E proof + stale-allowlist audit committed.
- Evidence: commit 9821e0f

§29. [ATM-029] Stale provider model-allowlists blocked current gemini/groq/zhipu models

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: hardcoded model allowlists predated current provider catalogs, rejecting valid current models; added current models, live-proven.
- Evidence: commit b23bcac

§30. [ATM-030] Coordination consensus tie-break was non-deterministic (map-iteration order)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: consensus tie-breaking iterated a map, producing non-deterministic winners across runs; made deterministic. Mutation-proven.
- Evidence: commit 77a0c15

§31. [ATM-031] Multipass verification chapter loop panicked when translation has fewer chapters than original

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the multipass loop indexed the translation by the original's chapter count, panicking when the translation had fewer chapters.
- Evidence: commit 9640a00

§32. [ATM-032] FB2 writer dropped section/subsection titles

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the FB2 writer omitted section and subsection <title> elements from the serialized output, losing chapter headings. Mutation-proven.
- Evidence: commit b83038c

§33. [ATM-033] PDF input revived — replaced license-gated unipdf with MIT ledongthuc/pdf

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: PDF input was dead behind the license-gated unipdf dependency; replaced with the MIT ledongthuc/pdf library. Real-translation proven.
- Evidence: commit 18d6f73

§34. [ATM-034] CLI ignored -o output format — always emitted EPUB

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the CLI hardcoded EPUB output regardless of the -o extension; now honors .txt/.fb2/.md/.epub (+ 2 §11.4.120 gate reconciliations).
- Evidence: commit ca5608f

§35. [ATM-035] Add .html/.htm CLI output format (completes the 5-output matrix)

Status: Implemented (→ Fixed.md) **Type:** Feature

- Summary: added HTML-escaped .html/.htm output, completing the output-format matrix. Real-translation proven.
- Evidence: commit fb07a59

§36. [ATM-036] DeepSeek model allowlist un-staled (accept deepseek-v4-flash/v4-pro)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the DeepSeek allowlist predated v4 models and rejected them; updated via live /models + real-translation verification (the canonical allowlist-refresh template).
- Evidence: commit 0fd1a34

§37. [ATM-037] WebSocket monitoring test harness data races + fatal map panic

Status: Fixed (→ Fixed.md) **Type:** Task

- Root cause: the monitoring test harness had concurrent map access causing data races and a fatal map-write panic; serialized access in the harness.
- Evidence: commit ab7db0e

§38. [ATM-038] TestSSHErrorHandling brittle error-string assertion (§11.4.1)

Status: Fixed (→ Fixed.md) **Type:** Task

- Root cause: the SSH error test asserted an exact error substring that varied by environment, producing FAIL-bluffs; assert on a stable condition instead.
- Evidence: commit 817b9dd

§39. [ATM-039] Preparation per-chapter analysis stamped no authoritative chapter number

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: per-chapter analysis records lacked the authoritative chapter number, breaking downstream attribution; now stamped.
- Evidence: commit 0df25d9

§40. [ATM-040] hardware parseLscpuCores undercounts physical cores on multi-socket hosts

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: parseLscpuCores counted cores per socket once instead of multiplying across sockets, undercounting on multi-socket machines and mistuning concurrency.
- Evidence: commit 80d627b

§41. [ATM-041] Batch same-stem/different-ext inputs collided onto one output file (data loss)

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: batch output naming used only the file stem, so a.txt and a.fb2 produced the same output path and overwrote each other.
- Evidence: commit c3117f8

§42. [ATM-042] Security RefreshToken resurrected expired/unbounded sessions into fresh valid tokens

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: token refresh did not re-check session expiry/bounds, so an expired session could be exchanged for a fresh valid token.
- Evidence: commit d0fe40a

**§43. [ATM-043] Empty/null LLMsVerifier envelope
({"models":null}) failed to decode, breaking /api/v1/verified-models**

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the verified-models decoder did not tolerate a null models array, so the endpoint broke whenever zero models were verified. Mutation-proven.
- Evidence: commit d8142e5

§44. [ATM-044] FB2 // dropped inline-element text

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the FB2 parser did not descend into inline elements inside <v>, <subtitle>, and <text-author>, dropping their text. Mutation-proven.
- Evidence: commit 8f52370

**§45. [ATM-045] Distributed fallback getFailureRate fabricated a
100% rate on window expiry / zero window**

Status: Fixed (→ Fixed.md) **Type:** Bug

- Root cause: the failure-rate calculation divided by an empty/expired window and reported a fabricated 100% failure rate, triggering needless fallback.
- Evidence: commit 20beda7

§46. [ATM-046] markdown→EPUB chapter